Project 1 STAT40850 Bayesian Analysis (online)

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The Dataset

```
A_alpha = 169
A_beta = 296-169
B_alpha = 247
B_beta = 380-247
```

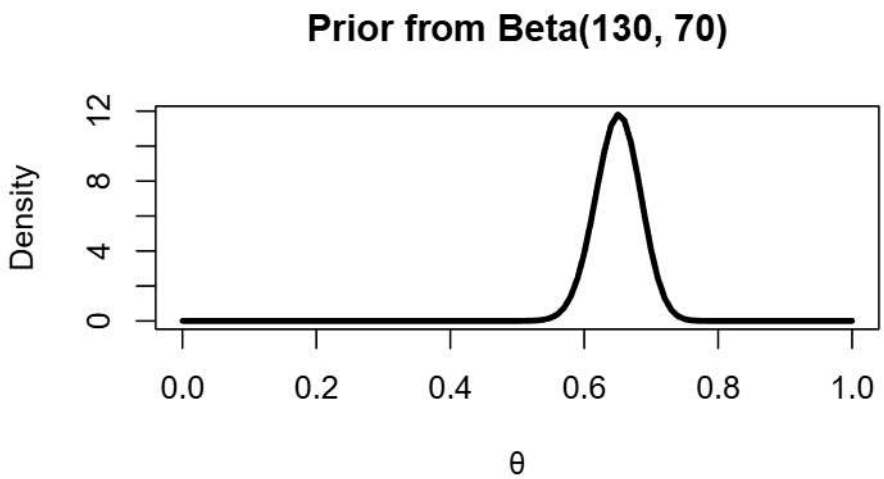
Question 1

```
#prior samples
A_s = 130
A_f = 200-130
#calculating the prior beta distribution with
#pr(thetaA <= 0.5)
shape1 = A_s
shape2 = A_f
#printing the probability of A <= 0.5.
cat("The probability of less or equal to 50% sucess rate last year
    \nwith method A is: ",
    pbeta(0.5, shape1, shape2, log = FALSE),
    "\nWith a center of", shape1/(shape1+shape2))
```

```
The probability of less or equal to 50% sucess rate last year

with method A is:  9.146204e-06
With a center of 0.65
```

```r
#Printing the cuve
curve(dbeta(x, A_s, A_f),
      main = "Prior from Beta(130, 70)",
      xlab = expression(theta),
      ylab = "Density",
      from = 0, to = 1,
      lwd = 3)
```



The center is /(+)=130/(130+70)=0.65. This means that, a priori, the expected pass rate is around 65%, and values equal or below t 50% are highly improbable. The prior probability $\Pr(\theta_A < 0.5) = 9.14 * 10\hat{}(-6)$. Based on the graph, the curve is highly informative and the priot density is peaked at around θ 0.65.

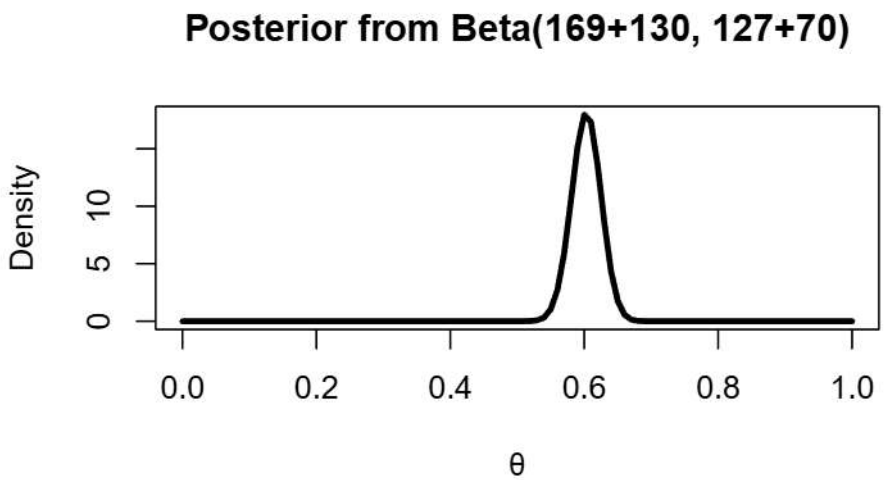
Question 2

```r
#calculationg the posterior beta distribution with
#pr(thetaA > 0.7)
shape1 = A_alpha+A_s
shape2 = A_beta+A_f

#printing and computiong the Pr(thetaA > 0.7 | xA)
cat('The posterior probability Pr(thetaA > 0.7 | xA) = ',
    pbeta(0.7, shape1, shape2, lower.tail = FALSE))
```

```
The posterior probability Pr(thetaA > 0.7 | xA) =  2.177698e-06
```

```r
#displaying the curve pf concentration for different theta.
curve(dbeta(x, shape1, shape2),
      main = "Posterior from Beta(169+130, 127+70)",
      xlab = expression(theta),
      ylab = "Density",
      from = 0, to = 1,
      lwd = 3)
```



The posterior probability $\Pr(\theta_A > 0.7 \mid \text{xA}) = 2.177698\text{e-}06$. We can see a strongly concentrated posterior distribution with the center being somewhere around Posterior mean 0.602. The posterior mass lies within approximately from $\theta = 0.5$ to $\theta = 0.7$ with very little density outside that range.

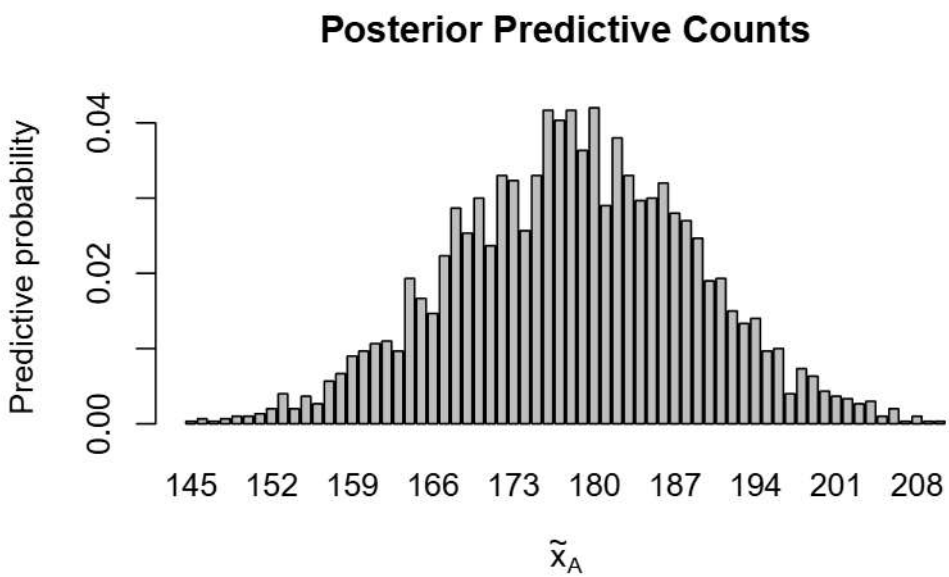
Question 3

We have seen that the probability of θ >0.7 is extremely small, which is a fact that perfectly aligns with the concentration of the posterior density in that θ. Therefore, we have evidence against the hypothesis that $\theta > 0.7$ a posteriori. Based on that information and the curve, the density reaches its highest point when the values of θ are near 0.6 (a point very close to the center of the curve), which makes θ 0.6 the most plausible.

Question 4

```
n_sim = 3000
set.seed(123)
#theta from posterior Beta(299, 197)
theta_s = rbeta(n_sim, 299, 197)
#Future counts from Binomial(296, theta)
x_sim = rbinom(n_sim, 296, theta_s)

#Barplot
barplot(table(x_sim)/sum(table(x_sim)),
        main = 'Posterior Predictive Counts',
        ylab='Predictive probability',
        xlab = expression(tilde(x)[A]))
```




```
#Mean - predictive probability
cat("P(x̃A >= 180 | xA ) =", mean(x_sim>= 180))
```

```
P(x̃A >= 180 | xA ) = 0.4533333
```

The posterior predictive distribution is centered near 178, slightly above the observed value $x_A = 169$. This is a reasonable fit for the model to the data. The estimated probability

$P(\tilde{x}_A \geq 180)$ 0.4533. This shows that observing 180 or more passes in a future set is fairly probable under the fitted model. That can be seen in the bar-plot also, there is a marked amount of concentration in posterior predictive mass close to or above 180.

Question 5

```
Sys.setenv(R_MAKEVARS_USER = normalizePath("~/.R/Makevars.win", winslash = "/"))
library(rstan)
rstan_options(auto_write = TRUE)
#The Stan code for the model
stancode = '
data
{
  //variables for A
  int <lower=0> nA;
  int <lower=0, upper=nA> xA;
  //Varables for B
  int <lower=0> nB;
  int <lower=0, upper=nB> xB;
}
parameters
{
  //theta for A
  real <lower=0, upper=1> thetaA;
  //theta for B
  real <lower=0, upper=1> thetaB;

}
model
{
  thetaB ~ beta(2, 2); // model for B
  xB ~ binomial(nB,thetaB);

  thetaA ~ beta(130, 70);  // Model for A
  xA ~ binomial(nA, thetaA);
}
generated quantities
{
  real diff;
  diff = thetaB - thetaA;//the posterior distribution
//of the difference theta_diff=thetaB-thetaA
```

```r
}'
#adding the stan model into a variable
invisible(capture.output(
  BB_model <- rstan::stan_model(model_code = stancode),
  file = NULL
))

#giving numbers to the stan code variables
nA = 296 ; xA = A_alpha
nB = 380 ; xB = B_alpha


#running the model with iter 1500.
BB_posterior <- sampling(
  BB_model,
  iter = 1500,
  seed = 123,
  data = list(nA = nA, xA = xA,
              nB = nB, xB = xB),
  refresh = 0
)
#printing the model results.
print(BB_posterior, pars = c("thetaA", "thetaB", "diff"))
```

```
Inference for Stan model: anon_model.
4 chains, each with iter=1500; warmup=750; thin=1;
post-warmup draws per chain=750, total post-warmup draws=3000.

       mean se_mean   sd  2.5%  25%  50%  75% 97.5% n_eff Rhat
thetaA 0.60       0 0.02  0.56 0.59 0.60 0.62  0.64  2457    1
thetaB 0.65       0 0.02  0.60 0.63 0.65 0.66  0.70  2604    1
diff   0.05       0 0.03 -0.02 0.02 0.05 0.07  0.11  2357    1

Samples were drawn using NUTS(diag_e) at Wed Feb 11 16:37:24 2026.
For each parameter, n_eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor on split chains (at
convergence, Rhat=1).
```

Method B seems a little more effective, with an average pass rate of around 65% compared to method A which has an approximate 60% pass rate. In order to determine which method is better, it is important to be mentioned that Method B's higher pass rate is relative due to the

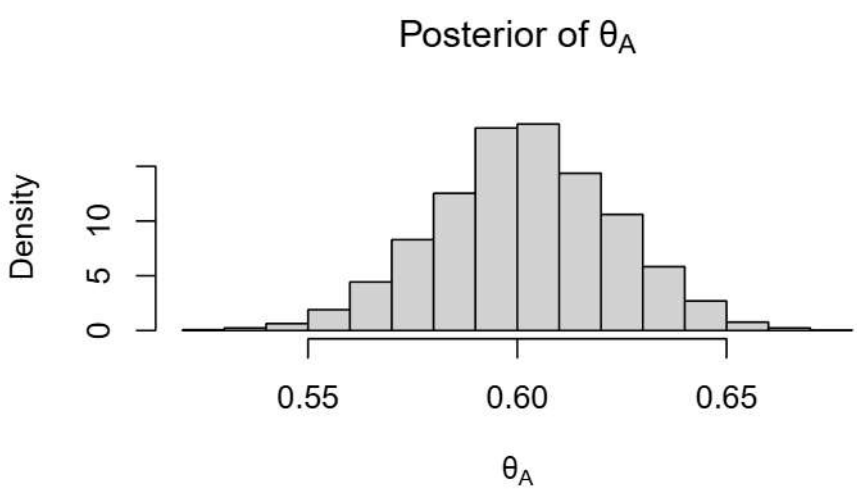
fact that the the difference θ_B - θ_A 0.05 and could be a result how the data was randomly generated. That indicates that we do not have enough evidence to be definitively sure that the method B provides for sure better results than method A.

Question 6

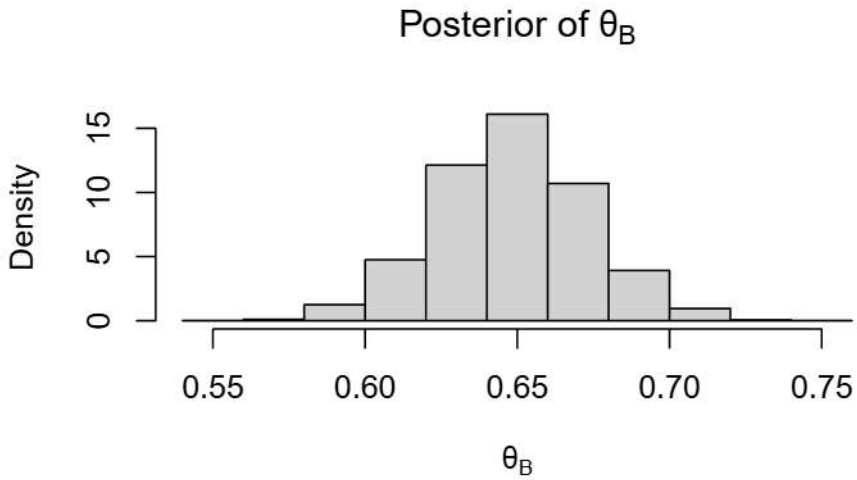
```
#Passing all sample values from the model as a matrix
pos_mat <- as.matrix(BB_posterior, pars = c("thetaA","thetaB","diff"))
dim(pos_mat)
```

```
[1] 3000    3
```

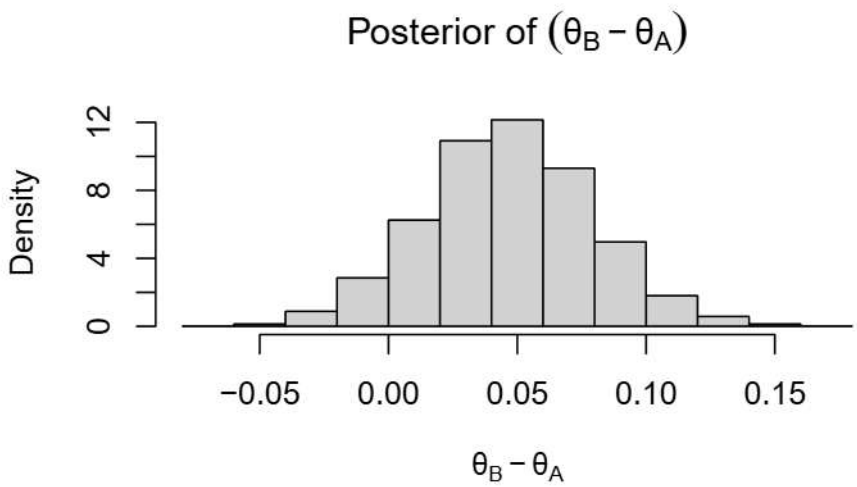
```
#histogram for thetaA
hist(pos_mat[, "thetaA"], probability = TRUE,
     main = expression("Posterior of " * theta[A]),
     xlab = expression(theta[A]))
```




```
#histogram for thetaB
hist(pos_mat[, "thetaB"], probability = TRUE,
     main = expression("Posterior of " * theta[B]),
     xlab = expression(theta[B]))
```




```
#histogram for thetaB-thetaA
hist(pos_mat[, "diff"], probability = TRUE,
     main = expression("Posterior of " * (theta[B] - theta[A])),
     xlab = expression(theta[B] - theta[A]))
```




```
#P(thetaA>thetaB|data)
Pdiffzero <- mean(pos_mat[, "diff"] < 0)
cat("P(theta_A > theta_B | data) =", Pdiffzero, "\n")
```

```
P(theta_A > theta_B | data) = 0.07766667
```

The distributions for the posteriors of θ_A and θ_B are both highly informative with a strongly distributed center. In case of posterior of θ_B, we observe the center of its concentration is a little higher θ_B 0.65 compared to the method A that the center is located around θ_A 0.60. For the posteriors of θ_B-θ_A, it is noticeable that we have a strong concentration at 0.05 which is the difference between the means of the posteriors θ_A and θ_B. The difference spans from -0.05 up to 0.15 with by far the most highly concentrated part be θ_B-θ_A 0.05 and the concentration clearly leans towards positive values. These results suggest results in favor of method B rather than method A, even though the conclusion can not be absolute.